

Turbulence kinetic energy

Modelling wind and dispersion in urban environments

Text book references: And Introduction to Boundary Meteorology, R.B. Stull, chapter 5

Learning Objectives:

After this class, you will be able to:

- Recognise the terms within the turbulent kinetic energy equations

Activities during the class:

During this class, you will perform the following activities:

1. Refreshing question
2. Find the equations terms
3. One-minute notes (the muddiest point)

**NO summative
assignments**

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Outline of the class:

1. Definition
2. TKE budget
3. Mean kinetic energy
4. Wrap-up

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Refreshing question

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Refreshing question

Find equations terms

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Definition

In the previous class we summarised the equations needed to forecast wind right, especially continuity and momentum:

$$\frac{\partial \overline{U_j}}{\partial x_j} = 0$$

$$\frac{\partial \overline{U_i}}{\partial t} + \overline{U_j} \frac{\partial \overline{U_i}}{\partial x_j} = -\delta_{i3} g + f_c \varepsilon_{ij3} \overline{U_j} - \frac{1}{\overline{\rho}} \frac{\partial \overline{P}}{\partial x_i} + \nu \frac{\partial^2 \overline{U_i}}{\partial x_j^2} - \frac{\partial (\overline{u'_i u'_j})}{\partial x_j}$$

These last term is a covariance, for which additional prognostic equations are required.

Turbulence departure of variables refers to the variables deviations from the means: $u', v', w', \dots$ in theory equations for these terms can be derived, with a bit of work

In this class, this is out of the scope, but these equations are directly linked to the turbulence kinetic energy concept, as you can see in the following turbulence kinetic energy budget equations

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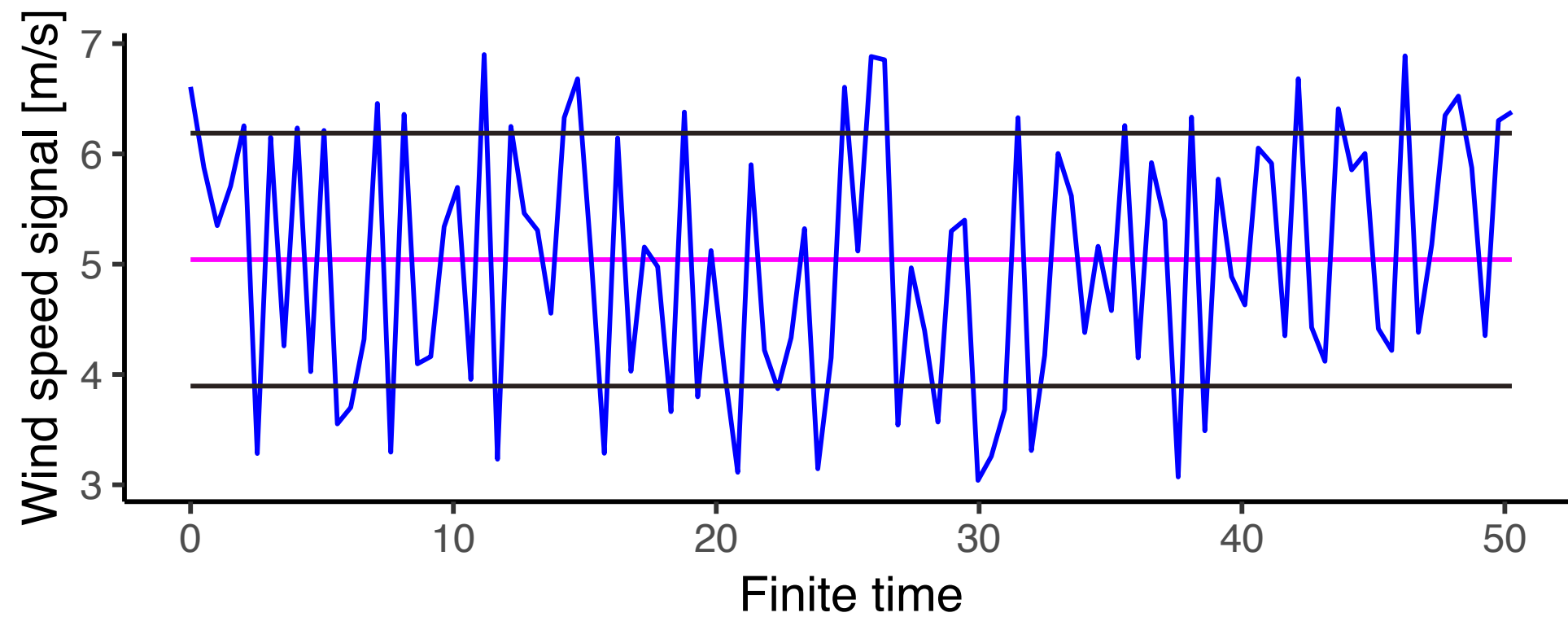
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Refreshing question

Refreshing question

Refreshing question



Definition

Turbulence kinetic energy (TKE) is one of the most important variables in micrometeorology, and therefore the flow around building environments.

- It measures the intensity of turbulence
- It is directly linked to the momentum, heat and moisture transport through the boundary layer
- Its individual equation terms describe physical processes that generate turbulence
—> the balance of those terms, will define if the flow is turbulent, or not

Its definition was already presented before:

$$\bar{e} = \frac{1}{2} \left(\overline{u'^2} + \overline{v'^2} + \overline{w'^2} \right) = \frac{1}{2} \overline{u_i'^2}$$

So if we have an equation for the velocity variances, we can get the TKE budget equation as we did for momentum and continuity

$$\frac{\partial \bar{e}}{\partial t} + \overline{U_j} \frac{\partial \bar{e}}{\partial x_j} = \delta_{i3} \frac{g}{\theta_v} \left(\overline{u_i' \theta_v'} \right) - \overline{u_i' u_j'} \frac{\partial \overline{U_i}}{\partial x_j} - \frac{\partial \left(\overline{u_j' e} \right)}{\partial x_j} - \frac{1}{\bar{\rho}} \frac{\partial \left(\overline{u_i' p'} \right)}{\partial x_i} - \varepsilon$$

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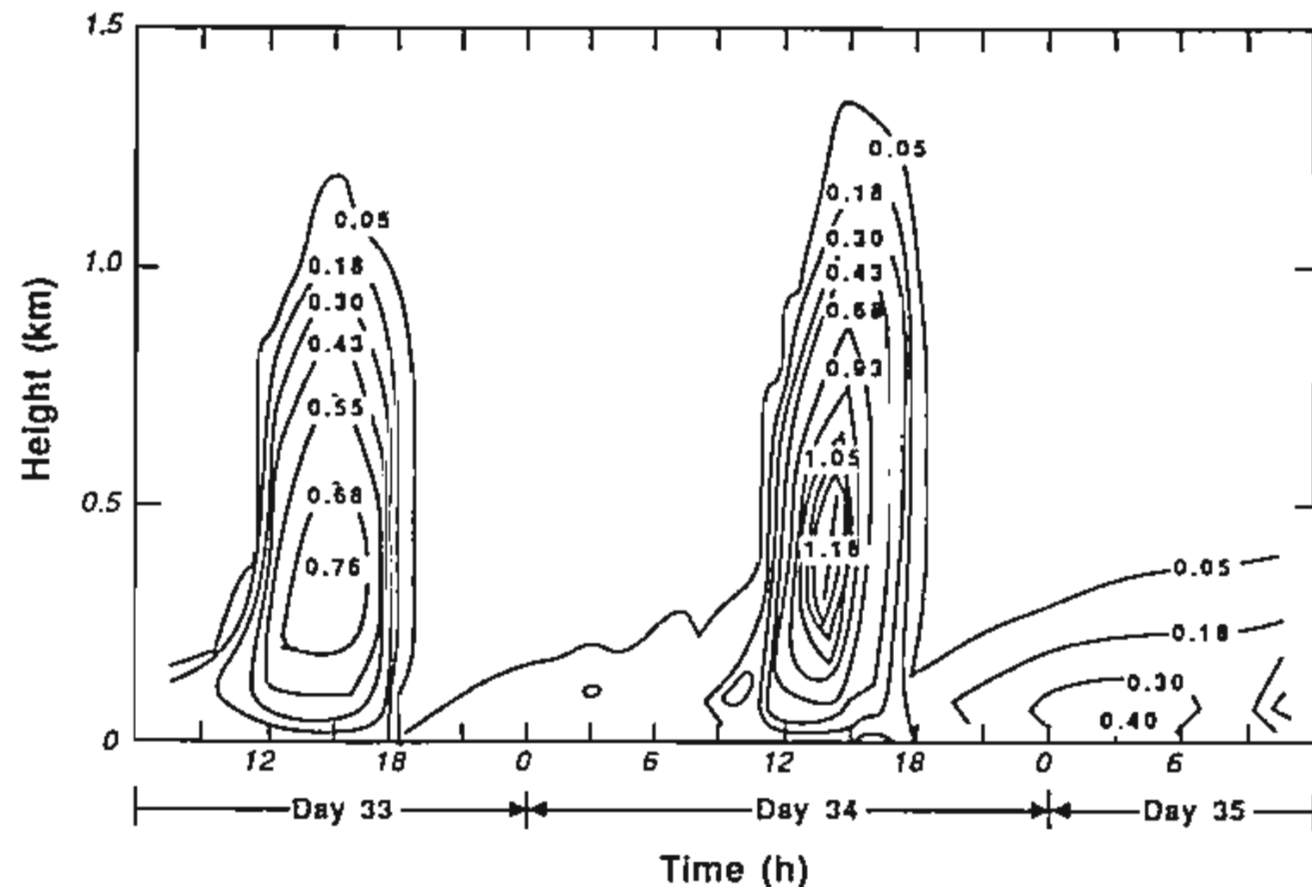
TKE budget

Each term describe physical processes that change turbulence and the balance of those terms, w define if the flow is turbulence:

$$\boxed{\frac{\partial \bar{e}}{\partial t}} + \bar{U}_j \frac{\partial \bar{e}}{\partial x_j} = \delta_{i3} \frac{g}{\theta_v} \left(\overline{u'_i \theta'_v} \right) - \overline{u'_i u'_j} \frac{\partial \bar{U}_i}{\partial x_j} - \frac{\partial \left(\overline{u'_j e} \right)}{\partial x_j} - \frac{1}{\bar{\rho}} \frac{\partial \left(\overline{u'_i p'} \right)}{\partial x_i} - \varepsilon$$

1. Storage or tendency

- Over the day, TKE can vary drastically with time and space
- Increase in TKE from a small early morning value to a larger early afternoon value represents a net storage of TKE in the air
- Non-turbulent FA air must be spun-up as entrainment incorporates into the mixed layer
- During late afternoon we have a negative storage term



TKE budget

Each term describe physical processes that generate turbulence and the balance of those terms, will define if the flow is turbulence:

$$\frac{\partial \bar{e}}{\partial t} + \boxed{\overline{U_j} \frac{\partial \bar{e}}{\partial x_j}} = \delta_{i3} \frac{g}{\theta_v} \left(\overline{u'_i \theta'_v} \right) - \overline{u'_i u'_j} \frac{\partial \bar{U}_i}{\partial x_j} - \frac{\partial \left(\overline{u'_j e} \right)}{\partial x_j} - \frac{1}{\bar{\rho}} \frac{\partial \left(\overline{u'_i p'} \right)}{\partial x_i} - \varepsilon$$

*2. Advection by mean
wind*

- Relatively little is known/understood about this term
- When averaged over a horizontal area larger than 10km, there is little variation of TKE
- This doesn't hold for urban environments
- This term is probably negligible over ocean surfaces
- Not negligible over water reservoirs or coastal areas: you can imagine a water reservoir colder than the surrounding land, where turbulence would decay in the overlaying air, while the air over the land is actively convective → a mean wind advecting air across the shores of the reservoir may cause significant change in this TKE budget

TKE budget

Each term describe physical processes that generate turbulence and the balance of those terms, will define if the flow is turbulence:

$$\frac{\partial \bar{e}}{\partial t} + \bar{U}_j \frac{\partial \bar{e}}{\partial x_j} = \boxed{\delta_{i3} \frac{g}{\theta_v} \left(\overline{u'_i \theta'_v} \right)} - \overline{u'_i u'_j} \frac{\partial \bar{U}_i}{\partial x_j} - \frac{\partial \left(\overline{u'_j e} \right)}{\partial x_j} - \frac{1}{\bar{\rho}} \frac{\partial \left(\overline{u'_i p'} \right)}{\partial x_i} - \varepsilon$$

*3. Buoyant production/
consumption*

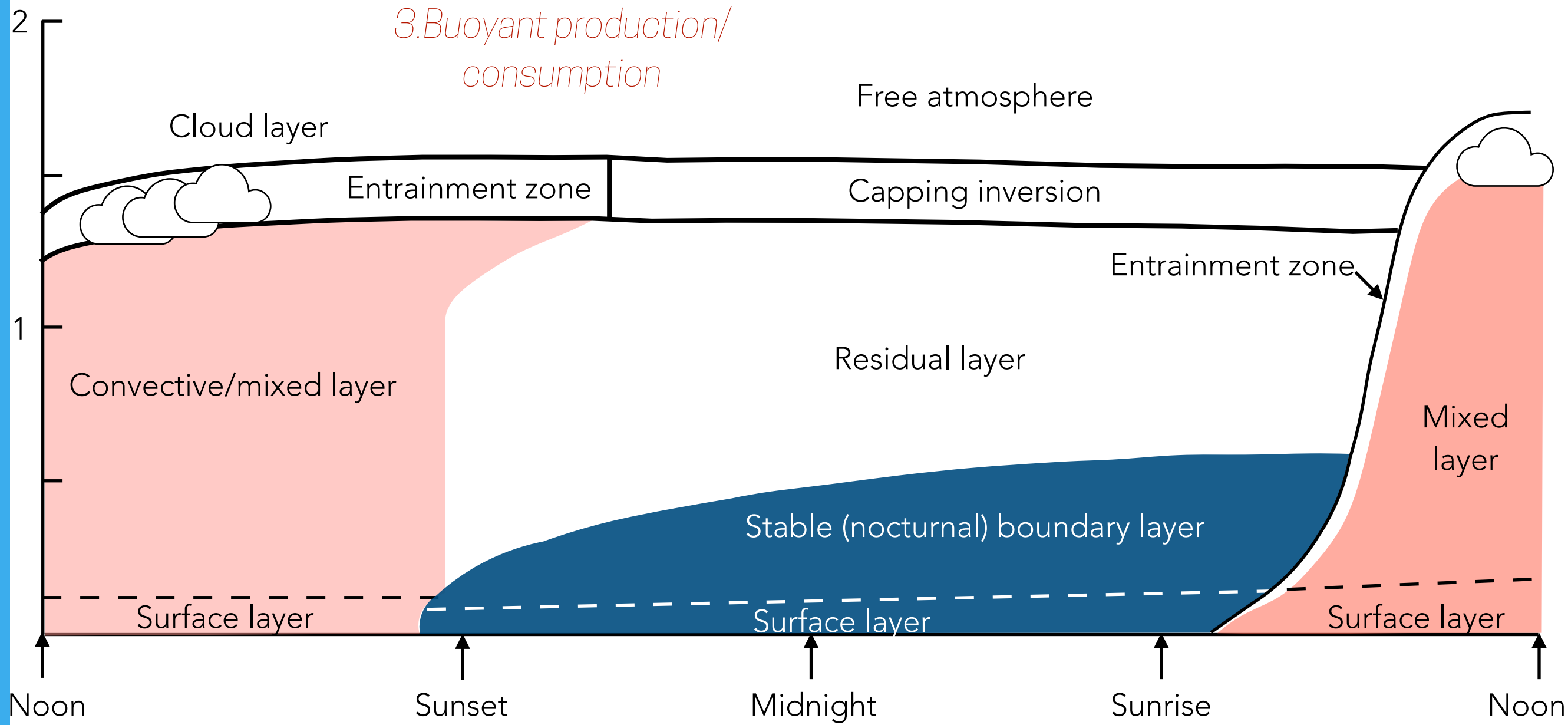
- The most important term is the flux of virtual potential temperature, which is positive and decreases roughly linearly with height within the bottom 2/3th of the convective ML
- Acts only in the vertical component —> anisotropic.
- **When positive:** represents the effects of thermals in the ML, values can go up to $1 \times 10^{-2} \text{m}^2/\text{s}^3$ near the ground on sunny days over land, it would be much smaller during cloudy days
- **When negative:** during stable conditions, which surpasses or consumes TKE. During t night over land, or anytime the surface is colder than the overlaying air. Also at the top of the mixed layer

TKE budget

Each term describe physical processes that changes turbulence and the balance of those terms, will define if the flow is turbulence:

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*3. Buoyant production/
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Questions so far?

TKE budget

Each term describe physical processes that changes turbulence and the balance of those terms, will define if the flow is turbulence:

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*4. Mechanical (shear)
production*

- This term is typically positive, since mean gradient and momentum flux are opposite sign
- It is often associated with surface layer: maximum production occurs close to the surface, where the wind shear is maximum
- Near 0 production in the ML above the surface layer, because changes in wind speed are small
- Anytime the ground is colder than the air, the shear term is the only term generating turbulence
- Greatest shears associated with the change of U and V components of mean wind with height —> anisotropic, and strongest in the horizontal

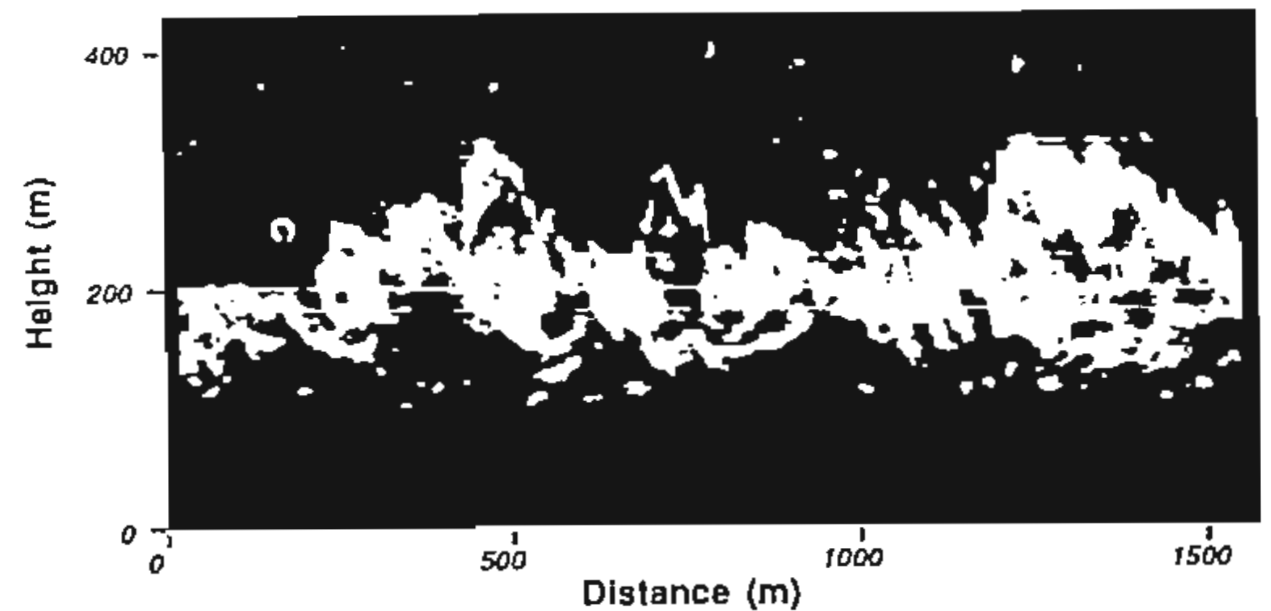
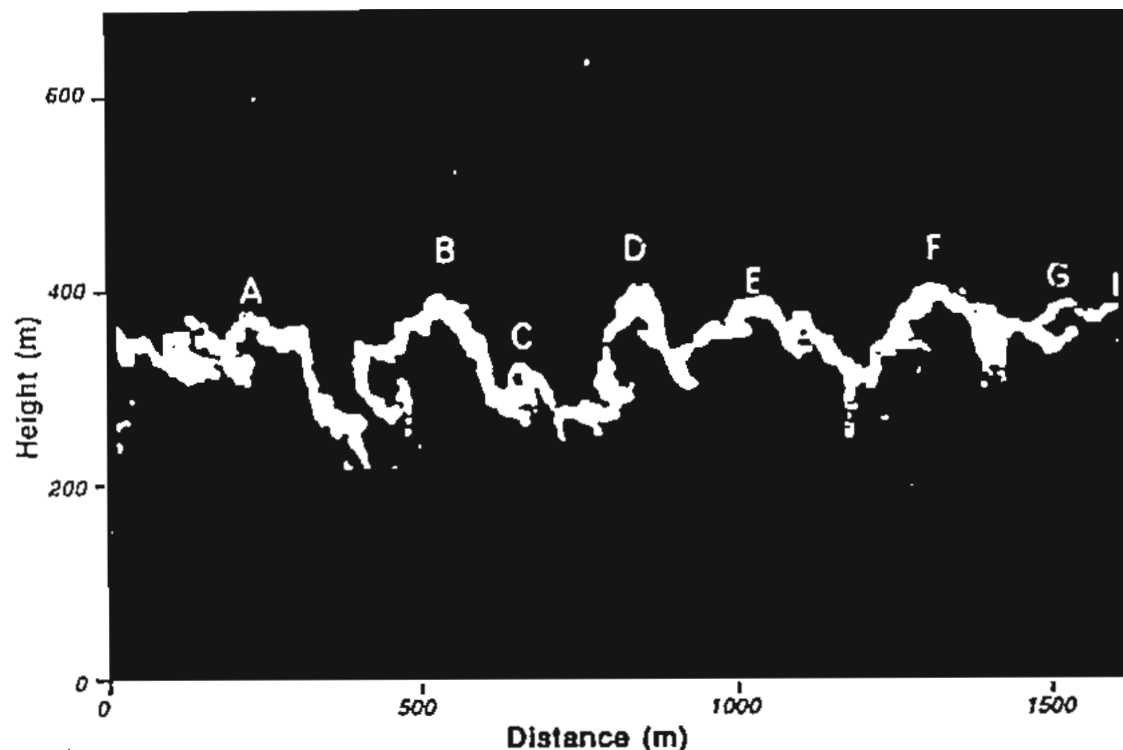
TKE budget

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4. Mechanical (shear)
production

- Differences between buoyancy (free convection) and shear (forced convection)



TKE budget

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5. Turbulent transport

- Neither creates, nor destroys TKE, but moves or redistributes from one location to another
- Integrated over the ML depth is identically to 0
- If there is more flux into a layer than leaves, TKE increases. On a local scale at any height of the ML, it can lead to production or loss of TKE
- w'^2 dominates in the middle of the ML and $(u'^2 + v'^2)$ dominates near the surface

Questions so far?

TKE budget

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6. Pressure correlation

- We know also few about this term, because measuring pressure fluctuations is very complicated, very small pressure fluctuations ~0.005-0.001kPa or less in the ML
- Normally this term is estimated as a residual from the budget, which means that accumulated errors in the other terms are an issue.
- Not only turbulence, but waves can also contribute to this term, and it is difficult to separate waves and turbulence within measurements (* see exceptional works).
- It does not only redistribute TKE, but it can draw energy out of the BL, ~<10% of total dissipation

* [There are some techniques used to separate waves and turbulence. For spectral waves i.e., wide bands, Madsen from MIT has proposed quite a few interesting ideas. For surface gravity waves, Bricker and Monismith (2007) suggest a wave separation technique that may be useful to apply.]

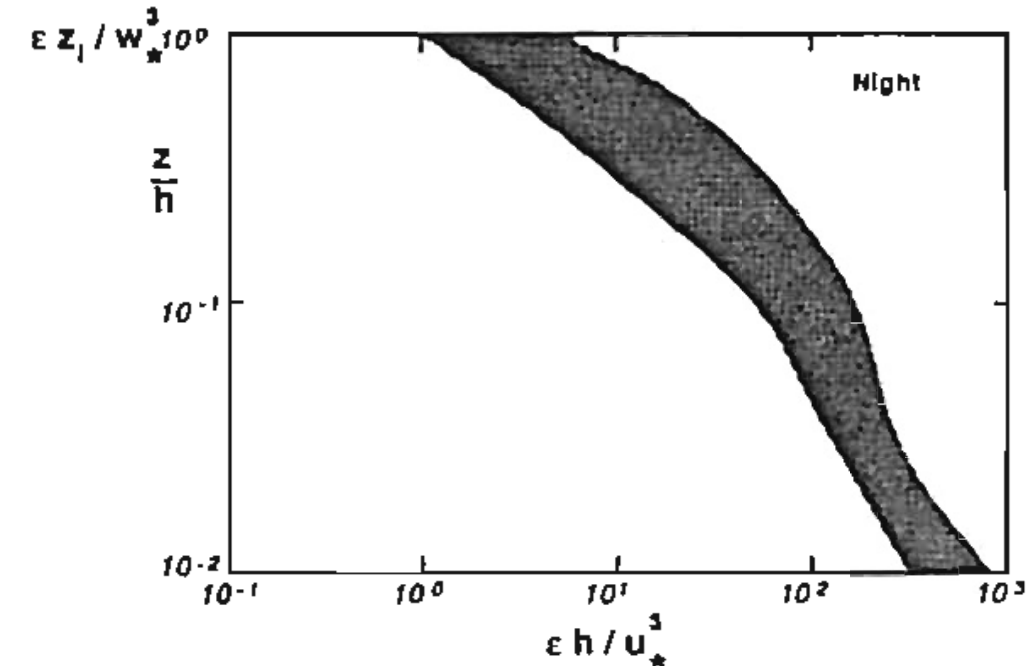
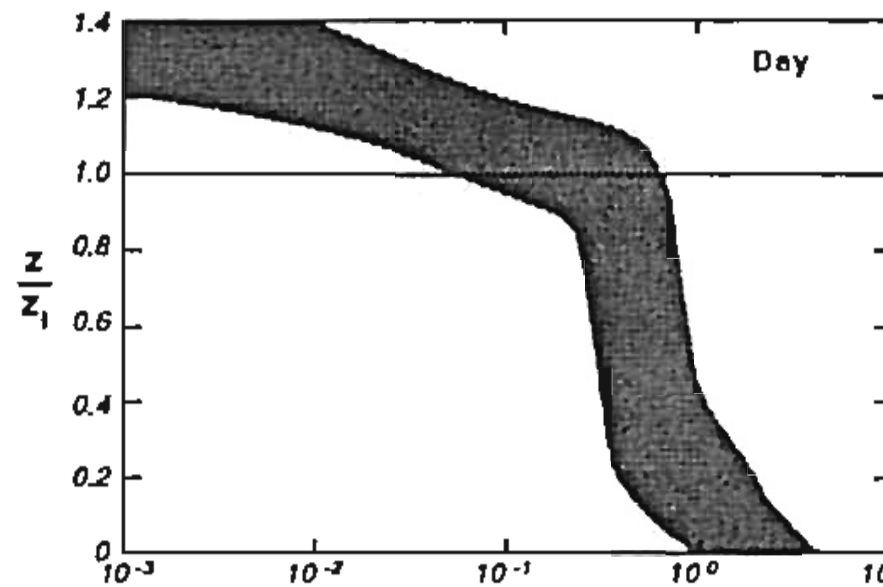
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- Molecular destruction of turbulent motions is greatest for the smallest size eddies. The more intense this small-scale turbulence, the greater the rate of dissipation
- Small scale turbulence is driven by the cascade of energy from larger scales
- **Turbulence is not conserved**, so the largest TKE and dissipation rates are found where production is largest (near the surface)
- Not expected to perfectly balance production because of transport terms

7. Dissipation



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Find equations terms

Finding the terms

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Turbulent
transport

$$\frac{\partial \bar{e}}{\partial t}$$

Storage or tendency

$$\delta_{i3} \frac{g}{\theta_v} \left(\overline{u'_i \theta'_v} \right)$$

Dissipation

$$-\overline{u'_i u'_j} \frac{\partial \bar{U}_i}{\partial x_j}$$

$$\bar{U}_j \frac{\partial \bar{e}}{\partial x_j}$$

Advection by mean
wind

$$-\frac{\partial \left(\overline{u'_j e} \right)}{\partial x_j}$$

Mechanical (shear)
production

$$-\frac{1}{\bar{\rho}} \frac{\partial \left(\overline{u'_i p} \right)}{\partial x_i}$$

$$-\varepsilon$$

Buoyant production/
consumption

Pressure
correlation

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Mean Kinetic Energy

In the TKE equation, 4th term involves the production of TKE by interaction of turbulence with the mean wind:

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Does this mean, that the production of TKE is accompanied by a corresponding loss of KE from the mean flow?

Obtaining the equation for the mean kinetic energy (MKE)

$$\frac{\partial \left(0.5 \bar{U}_i^2 \right)}{\partial t} + \bar{U}_j \frac{\partial \left(0.5 \bar{U}_i^2 \right)}{\partial x_j} = -g \delta_{i3} \bar{U}_i + f_c \varepsilon_{ij3} \bar{U}_i \bar{U}_j - \frac{\bar{U}_i}{\bar{\rho}} \frac{\partial \bar{P}}{\partial x_i} + \nu \bar{U}_i \frac{\partial^2 \bar{U}_i}{\partial x_j^2} + \boxed{\overline{u'_i u'_j} \frac{\partial \bar{U}_i}{\partial x_j}} - \frac{\left(\overline{u'_i u'_j} \bar{U}_i \right)}{\partial x_j}$$

YES, the energy that is mechanically produced from turbulence is lost from the mean flow

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